ANAEROBIC WORKSTATION

MAKE: BIOXIA



MODEL: A-300

BiOxia

www.bioxia.co.in

Construction & Design:

Durable, Non-Corrosive Acrylic Resin Sheet construction renders better visibility and insulation.

In built Digital Oxygen Display within the chamber indicate absence of Oxygen in the Workstation.

Dimensions (W × D × H, approx. in mm):

- External: 800 × 700 × 740
- Internal: 600 × 560 × 540
- Interlock: 110 × 180 × 180

Weight: Approx 80 Kg

Main Workstation Capacity

a. Up to 300 plates of 90mm × 15mm
b. Additional inoculation space for regular working.

Inter lock (Airlock) System Capacity

c. Integrated airlock supporting simultaneous transfer of at least 10 plates, with a rapid airlock cycle time of ≥ 20 seconds, along with provision for single-plate entry.
d. Auto Inter lock cycle

Interlock Cycle Time: 20 S

Atmosphere Control

Engineered to reliably maintain anoxic (oxygen-free) conditions for sensitive anaerobic applications.

Temperature Control

Heating element along with controller shall be provided to maintain the temperature inside chamber within 5°C above ambient up to 45°C, with Temperature Uniformity ± 2° C at 37°C

Temperature Controlled with Display

Gas Inlet & Supply Configuration

Inlet Options for Two Separate gases – one for premixed anaerobic gas & another one for Nitrogen gas

Gas Supply Configuration

- Compatible with operation using either:
  o One ~47-litre cylinder of standard anaerobic gas mixture (10% H_2, 10% CO_2, 80% N_2), or
  o One anaerobic gas mixture cylinder and one ~47-litre nitrogen cylinder.

Provision for Premixed anaerobic gas

To make the chamber oxygen-free, automatic forced amplifying purging system for Main Chamber as well as Interlock chamber, with adjustable purging period is provided

To make the chamber oxygen-free at faster rate, additional facility to condition the workstation atmosphere using Palladium Catalyst, is provided

Proper valve for purging, both for main chamber & Interlock chamber

Humidity Control

Capable of maintaining relative humidity levels of 80% RH or higher, subject to ambient operating conditions.
Humidity controlled with display

Heater for dehumidification

Oxygen & Catalyst Monitoring

Fully integrated anaerobic indicator and catalyst monitoring system, incorporating an O_2 sensor for continuous real-time oxygen level display.

Catalyst for O2 frees Atmosphere

Atmospheric conditioning system using Palladium Catalyst and Detox

Activated Detox (Charcoal) to absorb smell

User Interface & Monitoring – LCD Display

LCD Display with touch screen Controlled:

- Chamber pressure
- Temperature
- Fluorescent Light

Advanced touchscreen control panel enabling real-time monitoring of temperature, humidity, Gas pressure indication, Fluorescent Light and oxygen concentration level simultaneously.

To continuously monitor positive pressure inside the chamber Positive Pressure Display - Positive pressure (0.15 KPA/0.0015 bar) of anaerobic environment, continuously maintained inside the incubation chamber

Control Panel with Indicator

Low gas alarm with indicator on screen

Event Logging & Alarms

Comprehensive event logging system that records operational events with date and time stamps.

All alarms and system status notifications are clearly displayed on the touchscreen interface like visual alarm for gas cylinder pressure, chamber pressure and temperature.

Access & Handling Features

Leak proof arrangement from front-transparent-panel, for access inside chamber, for easy & comfortable handling/operation inside chamber

Access Ports with Easy Gloves Systems

Equipped with instant-access, sleeveless arm ports to ensure enhanced operator comfort, unrestricted hand movement, and convenient transfer of samples or bulk materials.

Condensate controller technology

The front 10% slope viewing panel

Utility & Control Features

Internal Power Supply

Provided with internal electrical sockets for convenient operation of accessories and instruments.

Utility port with “flexible hose” for manual purging by Nitrogen, inside the chamber.

Operated by foot switch outside chamber. Similar Foot Switch is provided for operation for gas nozzle ports purging & spot lamp inside chamber

Two Utility connection (Nozzles) ports

Auto excess pressure control system

Safety Features

Excess pressure relief valve

Temperature overshoot controller

Resettable Over current breaker fitted for protection

Internal Illumination

Fitted with a built-in inspection spot lamp and uniform internal lighting system for clear visibility of samples and workspace.

Interior Shelving

Supplied with a full-length internal shelf to maximize usable working and storage space.

Electrical Requirements

Power Supply : 230 V ± 15%, 50 Hz ± 3%

Supplied Complete With

- Anaerobic gas cylinders (~47 L) and matching pressure regulators.
- An electronic gas leak detection system

Certifications:

ISO 9001:2015 Certified

CE Marked : CE marked product

ISO 14001:2015

ISO 27001:2022

GMP Certified

Applicable Standards

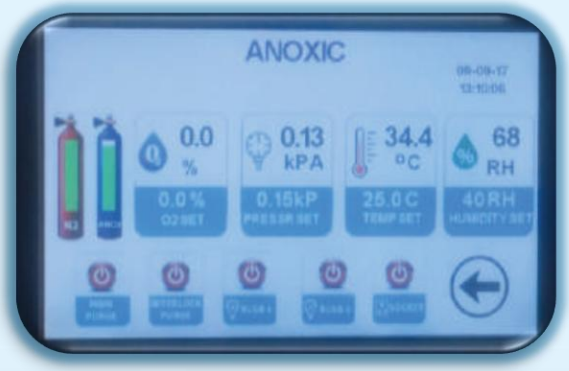
EN ISO 14971:2012, EN ISO 13485 : 2012, EN 980:2008, EN 1041 : 2008, EN 61010-1:2010
EN 61326-1:2013, EN 12469:2000, Device Directive 98/79/EC ,93/42/EEC, 73/23/EEC, 89/336/EEC and 89/392/EEC, EN IEC 61010-1:2010+A1:2019, EN 61326-1: 2021, EN 12469:2000, IEC 60601-1-11:2015

Calibration of Oxygen





Touch Screen Display



Data Logging Facility



Interlock Transfer



Condensing Cooling Plate



Settings



“Flexible hose” One Utility Connection Port



Spot Lamp



Foot Switch



Bare Hand Facility



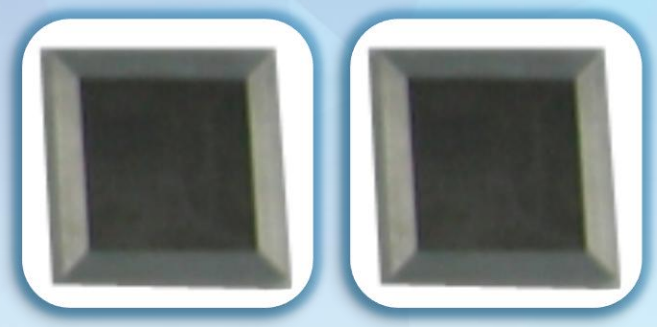
Petri Dish Holders



Gloves



Catalyst & Detox Pouch





PRODUCT SUMMARY OF WORKSTATION

MODEL	A300
Max Plate Capacity	300
Interlock Plate Capacity	10
Interlock Cycle Time	20 Second
Cabinet Mode of Construction	Acrylic Resin Sheet
Max Incubation Temperature	45° C
Gas Supply	10% CO_2 / 10% H_2 / 80% N_2 anaerobic gas mixture, and Nitrogen
Humidity control	√
Condensate Control	√
Positive Pressure Display	√
Safety Excess Pressure Relief Valve	√
Low Gas Alarm	√
Inspection Spot Lamp	√
Fluorescent Illumination	√
Palladium Catalyst	√
Activated Carbon Particles	√
Oxygen Sensor with Display	√
Oxygen Controlled with Display 0.0% to 22.00%	√
Ultra Violet (UV) Light	O
Bare Hand Facility	√
Incubator	O
Foot Switch Operation	√
Removable Front	O
Internal Power Supply	√
Utility nozzle Connection	√
LCD Display with control	√
Fan Unit	√
Adjustable Shelf	√
Cabinet Stand	O
Regulator for Cylinder	√
Gloves	√
Vacuum Line Port	√
Cable Gland Port	√
Petri Dish Holder	√

√ - Available O - Optional N/A - Not Available